

Pure Mathematics 1

Topic 6 – Trigonometric Ratios

**Exam Questions
2019 – 2024**

Marks: / 127

Exam Questions

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5.

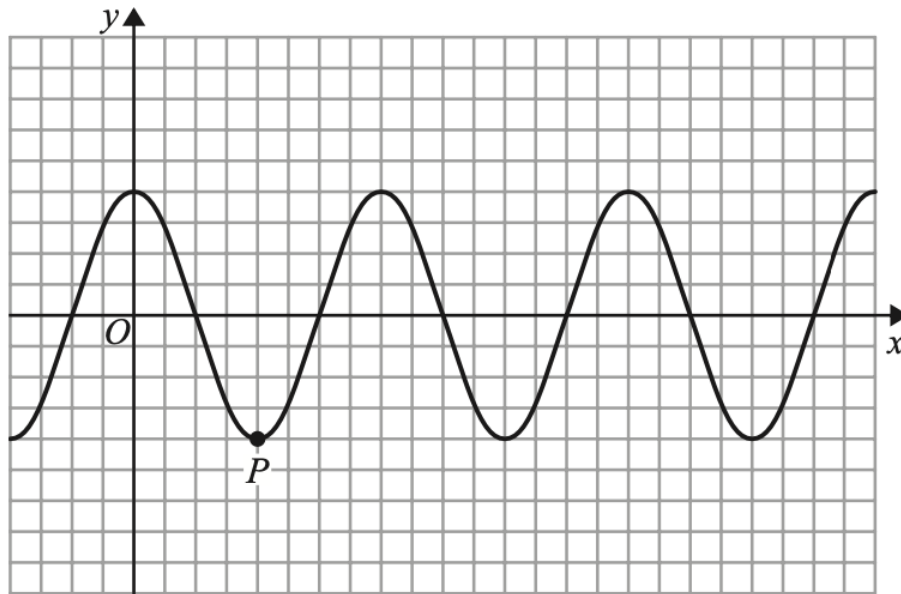


Figure 2

Figure 2 shows a plot of part of the curve with equation $y = \cos 2x$ with x being measured in radians.

The point P , shown on Figure 2, is a minimum point on the curve.

(a) State the coordinates of P .

(2)

A copy of Figure 2, called Diagram 1, is shown at the top of the next page.

(b) Sketch, on Diagram 1, the curve with equation $y = \sin x$

(2)

(c) Hence, or otherwise, deduce the number of solutions of the equation

(i) $\cos 2x = \sin x$ that lie in the region $0 \leq x \leq 20\pi$

(ii) $\cos 2x = \sin x$ that lie in the region $0 \leq x \leq 21\pi$

(2)

Question 5 continued

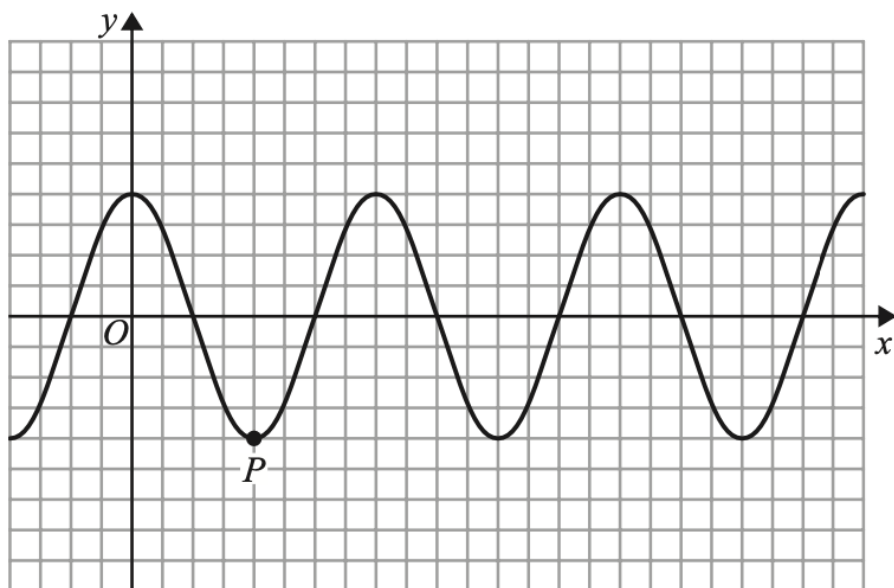
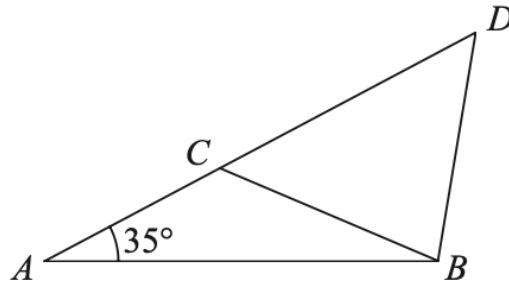


Diagram 1

2019 January Q7 P1

7.



Not to scale

Figure 3

Figure 3 shows the design for a structure used to support a roof.

The structure consists of four wooden beams, AB , BD , BC and AD .

Given $AB = 6.5$ m, $BC = BD = 4.7$ m and angle $BAC = 35^\circ$

- (a) find, to one decimal place, the size of angle ACB ,
(3)
- (b) find, to the nearest metre, the total length of wood required to make this structure.
(3)

2019 June Q9 P1

9.

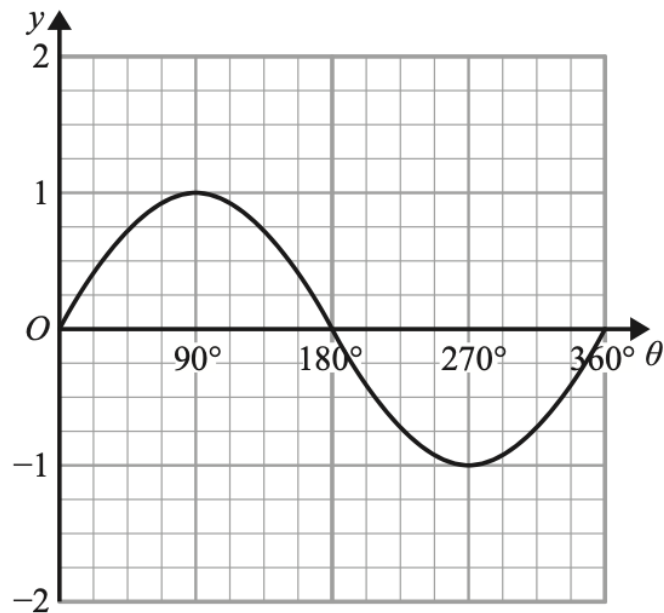


Figure 3

Figure 3 shows a plot of the curve with equation $y = \sin \theta$, $0 \leq \theta \leq 360^\circ$

(a) State the coordinates of the minimum point on the curve with equation

$$y = 4 \sin \theta, \quad 0 \leq \theta \leq 360^\circ$$

(2)

A copy of Figure 3, called Diagram 1, is shown on the next page.

(b) On Diagram 1, sketch and label the curves

(i) $y = 1 + \sin \theta$, $0 \leq \theta \leq 360^\circ$

(ii) $y = \tan \theta$, $0 \leq \theta \leq 360^\circ$

(2)

(c) Hence find the number of solutions of the equation

(i) $\tan \theta = 1 + \sin \theta$ that lie in the region $0 \leq \theta \leq 2160^\circ$

(ii) $\tan \theta = 1 + \sin \theta$ that lie in the region $0 \leq \theta \leq 1980^\circ$

(3)

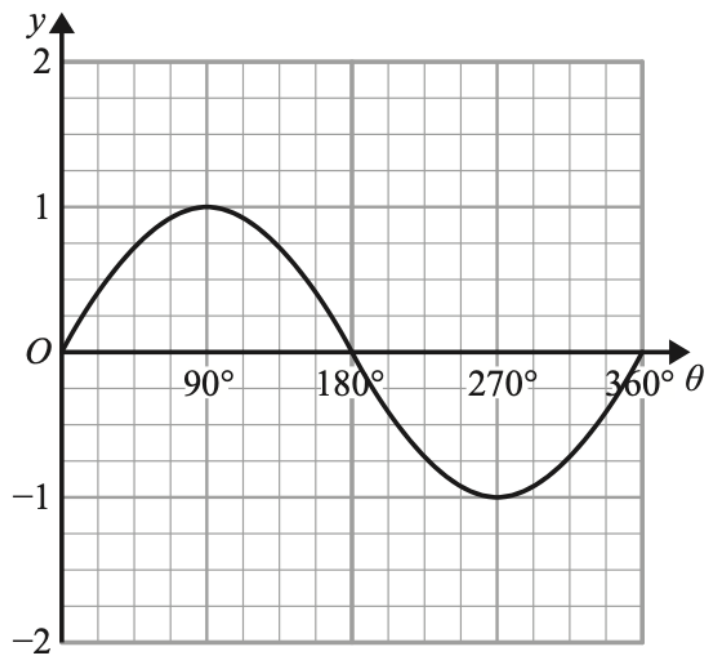


Diagram 1

2019 October Q4 P1

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

- (a) the size of angle DAB , in degrees, to 2 decimal places,

(3)

- (b) the length of diagonal BD , in cm , to one decimal place.

(2)

9.

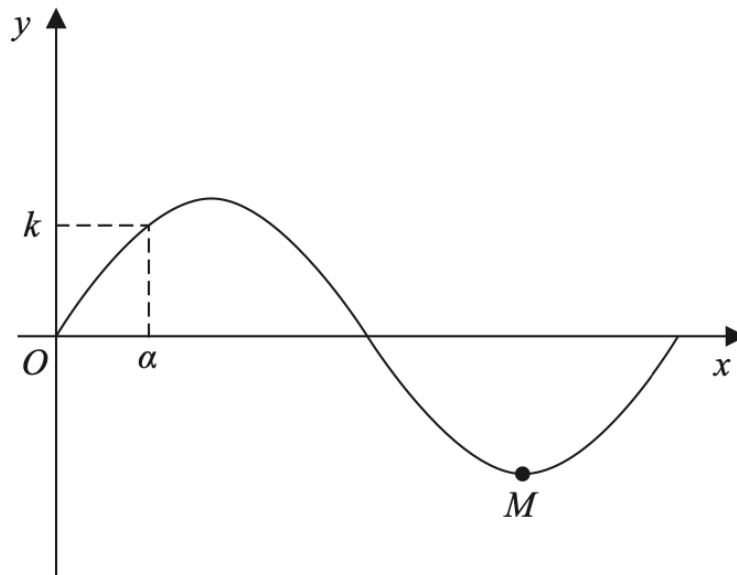


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

(a) State the period of C . (1)

(b) State the coordinates of M . (1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,
 (ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$. (2)

7.

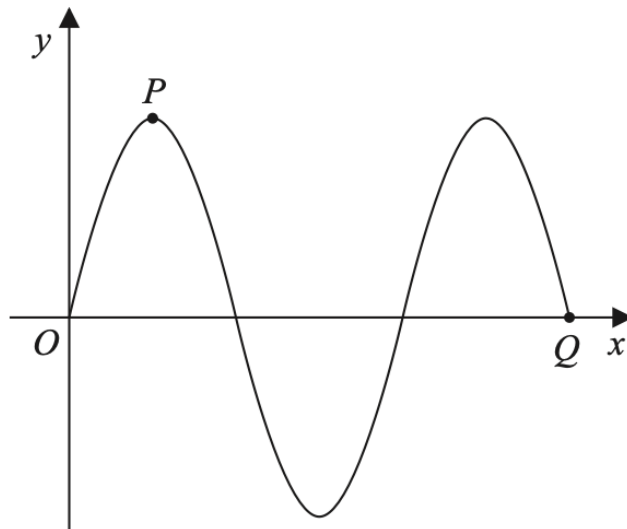


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

3.

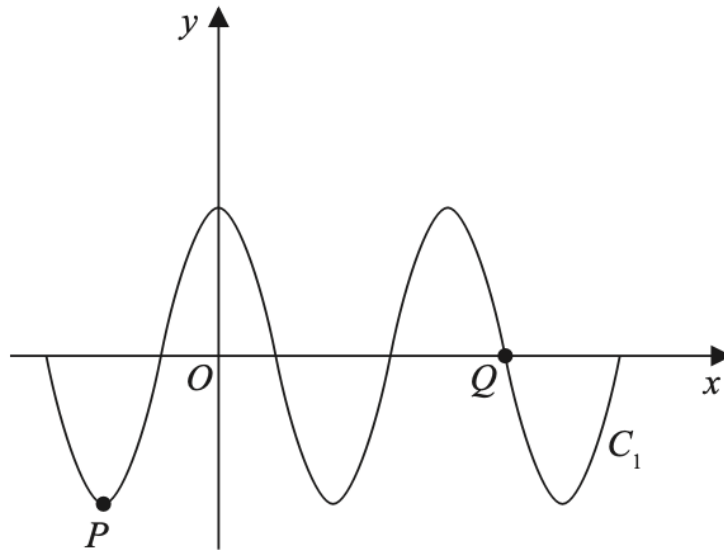


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4 \cos x^\circ$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4 \cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

2021 June Q3 P1

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

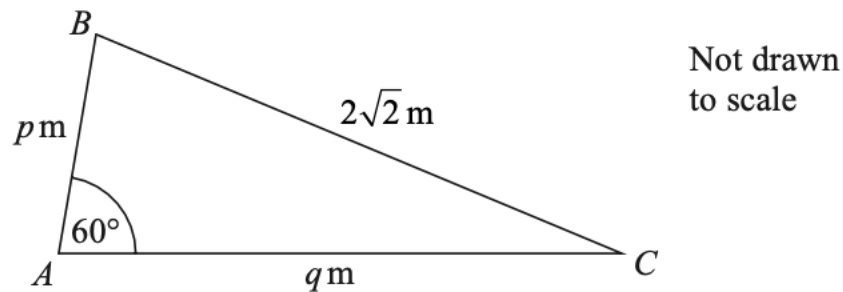


Figure 1

Figure 1 shows the plan view of a flower bed.
The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

(a) Show that

$$p^2 + q^2 - pq = 8 \quad (2)$$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
- (ii) the exact value of q . (5)

Using the answers to part (b),

- (c) calculate the exact area of the flower bed. (2)

9.

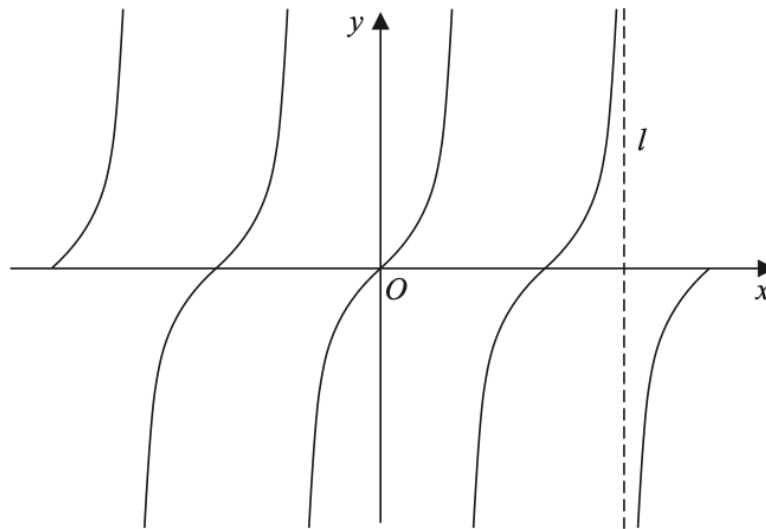


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

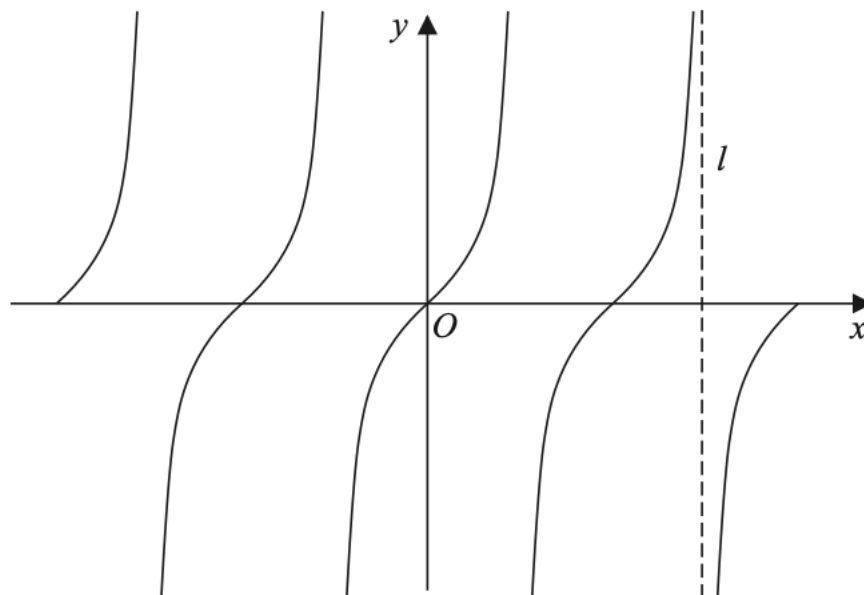


Diagram 1

4.

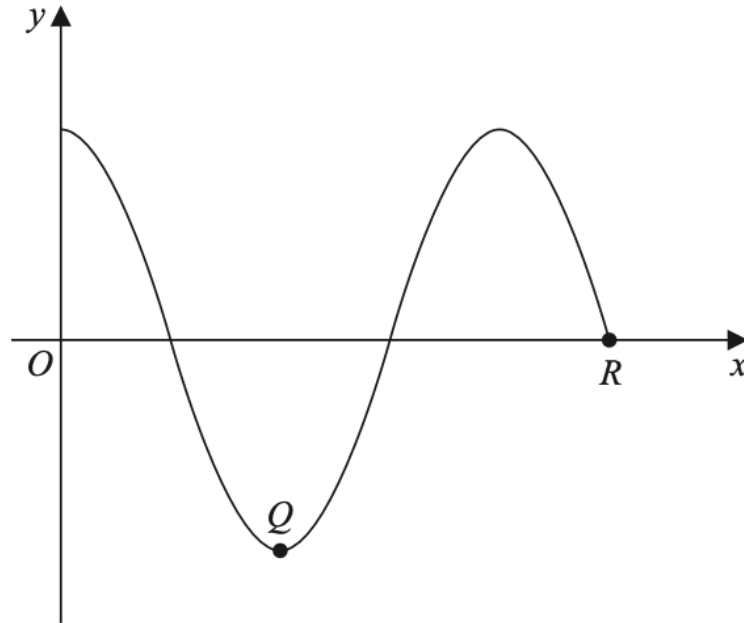


Figure 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x^\circ \quad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

- (i) the coordinates of Q ,
- (ii) the value of k .

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)

9.

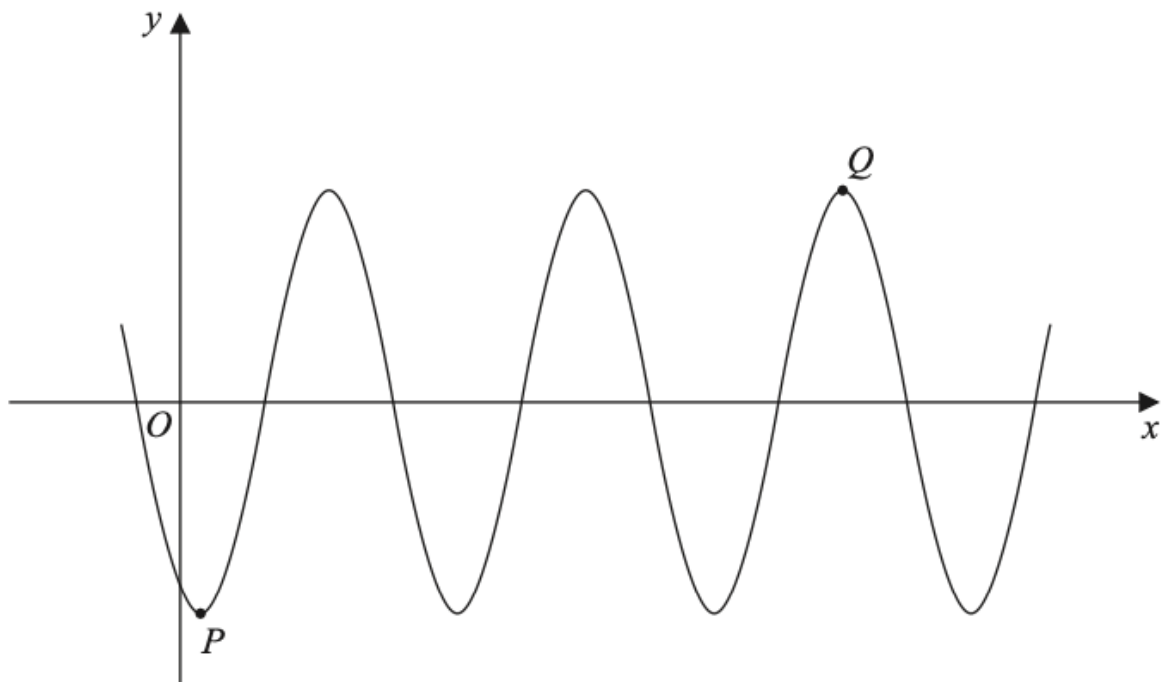


Figure 4

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A .

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)

2022 June Q2 P1

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places. (2)

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places. (3)

9.

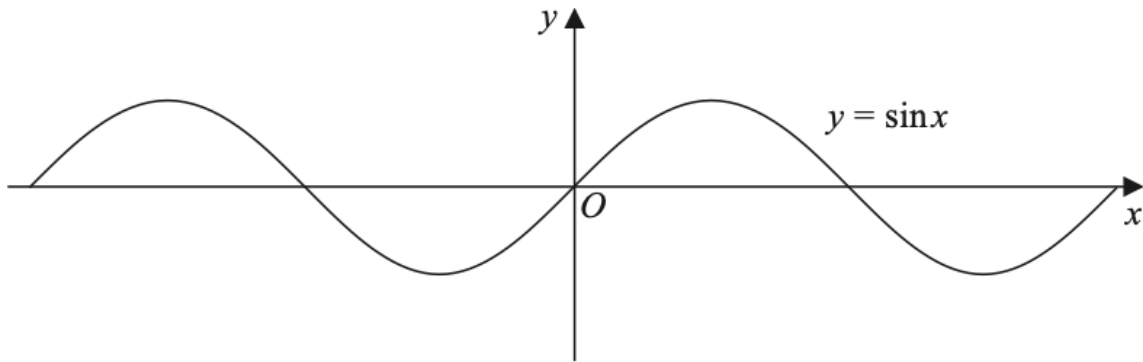


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

(a) state, in terms of p , the value of

(i) $2 \sin(180^\circ - \alpha)$

(ii) $\sin(\alpha - 180^\circ)$

(iii) $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

Question 9 continued

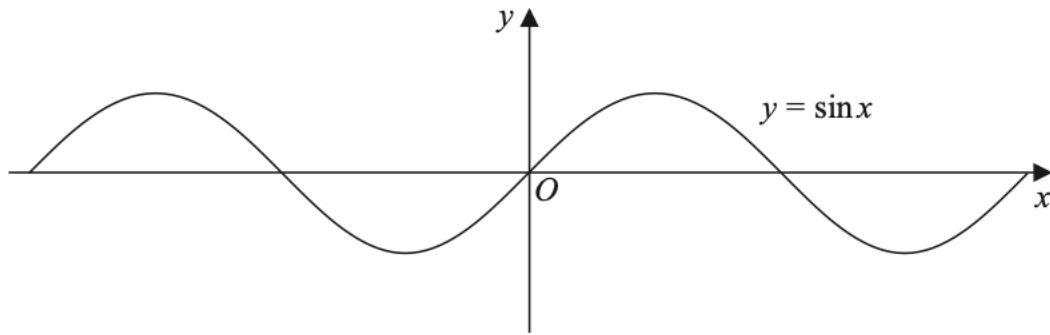


Diagram 1

9.

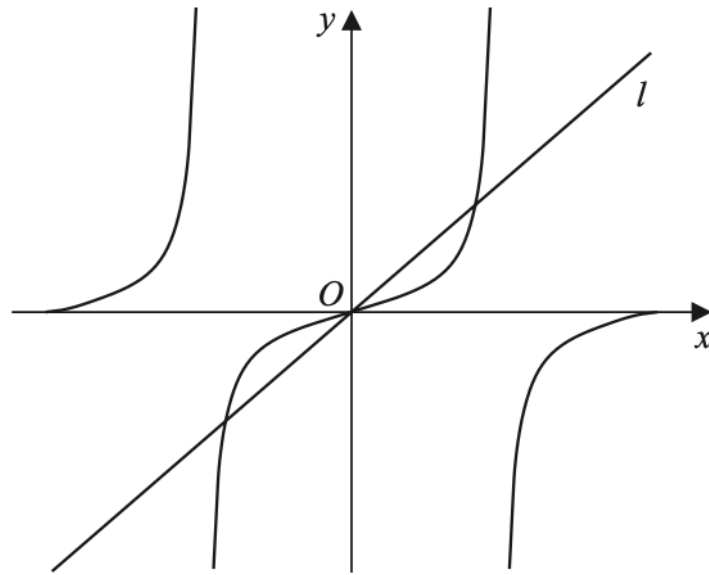


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

9. (i)

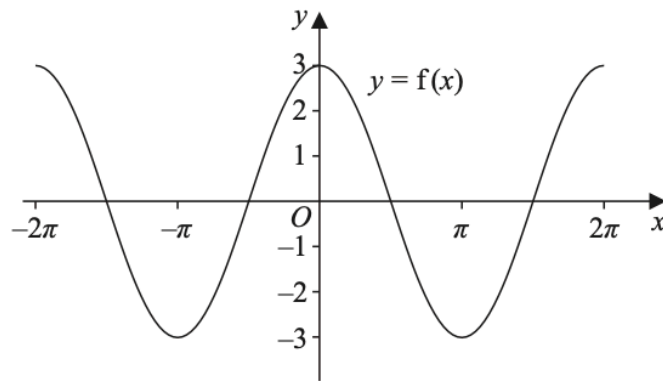


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

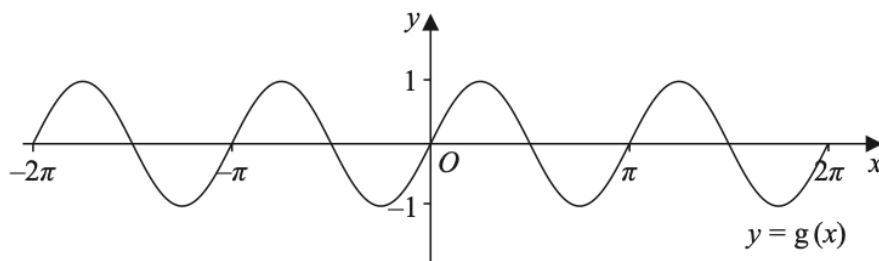


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

2023 October Q5 P1

5.

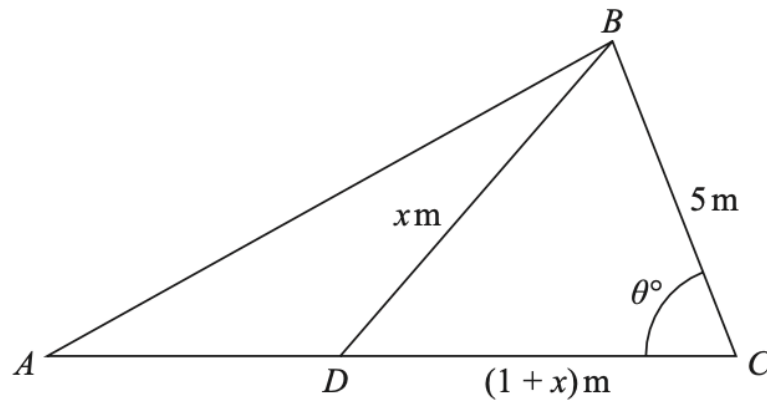


Diagram **NOT**
accurately drawn

Figure 2

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

- $BD = x \text{ m}$
- $CD = (1 + x) \text{ m}$
- $BC = 5 \text{ m}$
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30^\circ$
- ADC is a straight line

(b) find the area of triangle ABC , giving your answer, in m^2 , to one decimal place.

(5)

10.

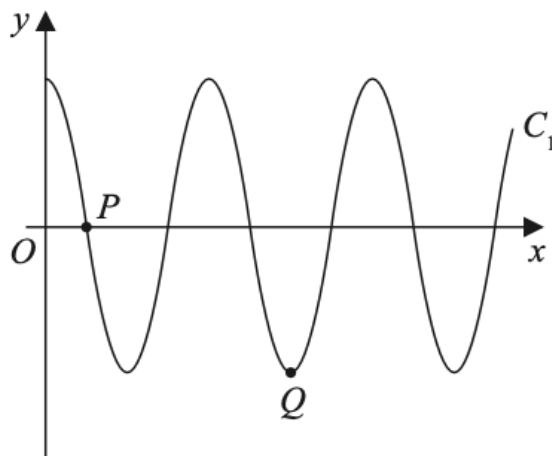


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

2024 January Q2 P1

2. The triangle ABC is such that

- $AB = 15 \text{ cm}$
- $AC = 25 \text{ cm}$
- angle $BAC = \theta^\circ$
- area triangle $ABC = 100 \text{ cm}^2$

(a) Find the value of $\sin \theta^\circ$

(2)

Given that $\theta > 90$

(b) find the length of BC , in cm, to 3 significant figures.

(3)

2024 January Q6 P1

6.

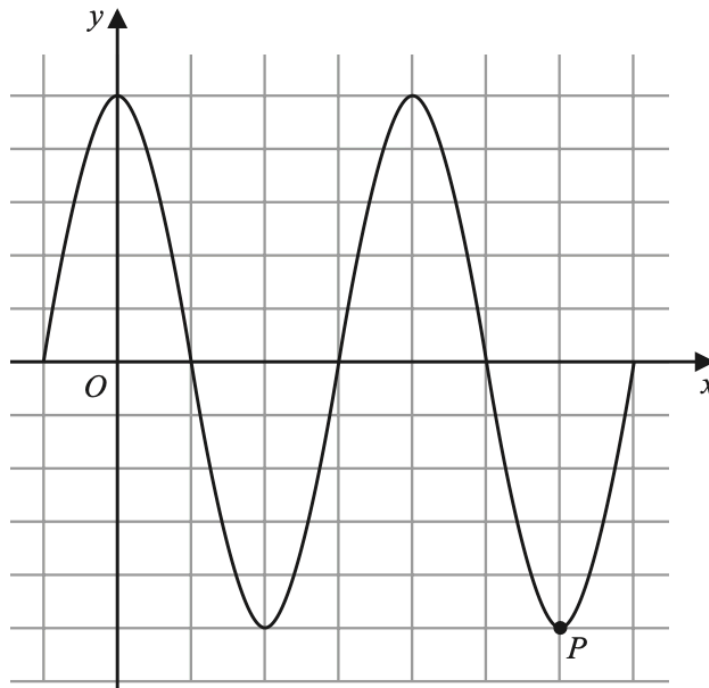


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

11.

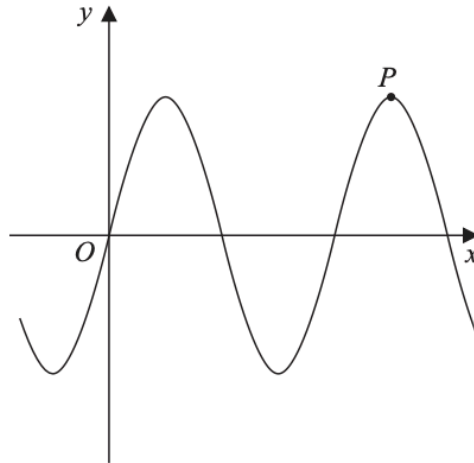


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

9.

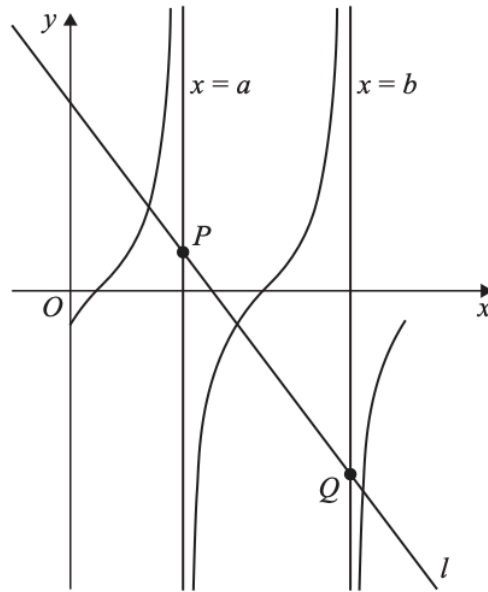


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan\left(x - \frac{\pi}{6}\right)$ for $0 \leq x \leq 2\pi$
- part of the straight line l with equation $y = \pi - x$

(a) State the number of solutions of the equation

- $\tan\left(x - \frac{\pi}{6}\right) = \pi - x$ in the interval $0 \leq x \leq 2\pi$
- $\tan\left(x - \frac{\pi}{6}\right) = \pi - x$ in the interval $0 \leq x \leq 100\pi$
- $\tan\left(x - \frac{\pi}{6}\right) = \pi + x$ in the interval $0 \leq x \leq 2\pi$

(3)

The line with equation $x = a$, shown in Figure 3, is the asymptote to the curve with the smallest positive x coordinate.

(b) State the value of a

(1)

The line with equation $x = b$, also shown in Figure 3, is the asymptote to the curve with the second smallest positive x coordinate.

The line l meets $x = a$ at point P and meets $x = b$ at point Q as shown in Figure 3.

(c) Find the midpoint of the line segment PQ .

(4)

Pure Mathematics 1

Topic 7 – Radians

**Exam Questions
2019 – 2024**

Marks: / 153

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2019 January Q10 P1

10. A sector AOB , of a circle centre O , has radius r cm and angle θ radians.

Given that the area of the sector is 6 cm^2 and that the perimeter of the sector is 10 cm ,

(a) show that

$$3\theta^2 - 13\theta + 12 = 0$$

(4)

(b) Hence find possible values of r and θ .

(3)

7.

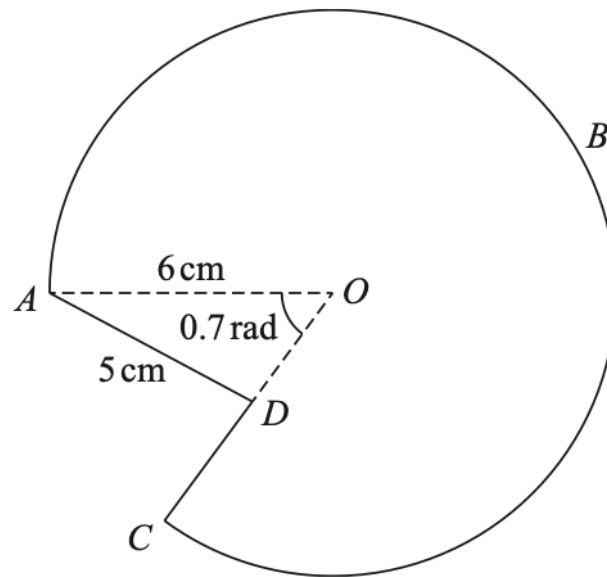


Figure 2

The shape $ABCD$ consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD , as shown in Figure 2.

The point D lies on OC .

The radius of the circle is 6 cm, length AD is 5 cm and angle AOD is 0.7 radians.

(a) Find the area of the sector $ABCOA$, giving your answer to one decimal place.

(3)

Given angle ADO is obtuse,

(b) find the size of angle ADO , giving your answer to 3 decimal places.

(3)

(c) Hence find the perimeter of shape $ABCD$, giving your answer to one decimal place.

(4)

2019 October Q1 P1

1.

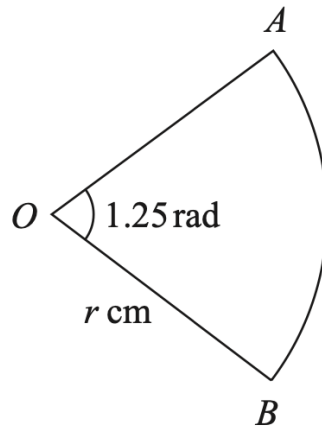


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

(a) find the exact value of r ,

(2)

(b) find the exact length of the perimeter of the sector. Write your answer in simplest form.

(3)

4.

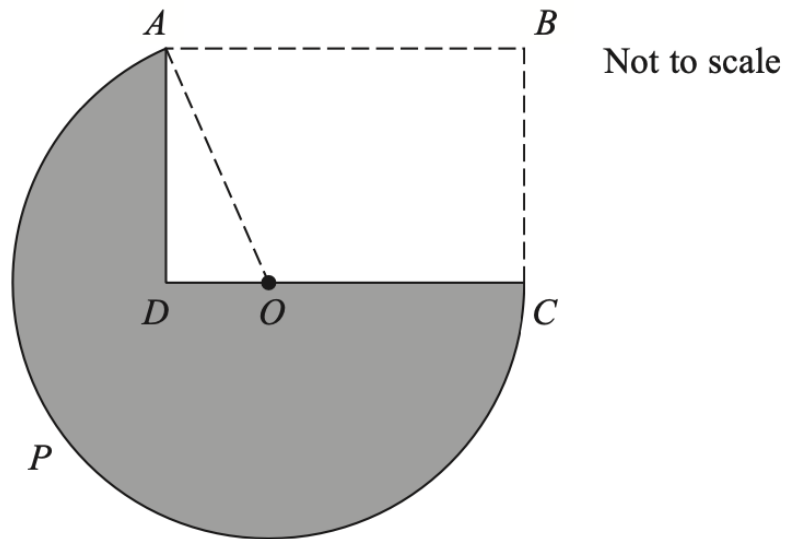


Figure 2

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$ABCD$ is a rectangle with $AB = 16$ m.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The point O lies on the line DC , as shown in Figure 2.

(a) Show that the size of angle AOD is 1.231 radians to 3 decimal places.

(2)

The lawn $APCDA$ is shown shaded in Figure 2.

(b) Find the area of the lawn, in m^2 , to one decimal place.

(4)

(c) Find the perimeter of the lawn, in metres, to one decimal place.

(3)

3.

Diagram not
drawn to scale

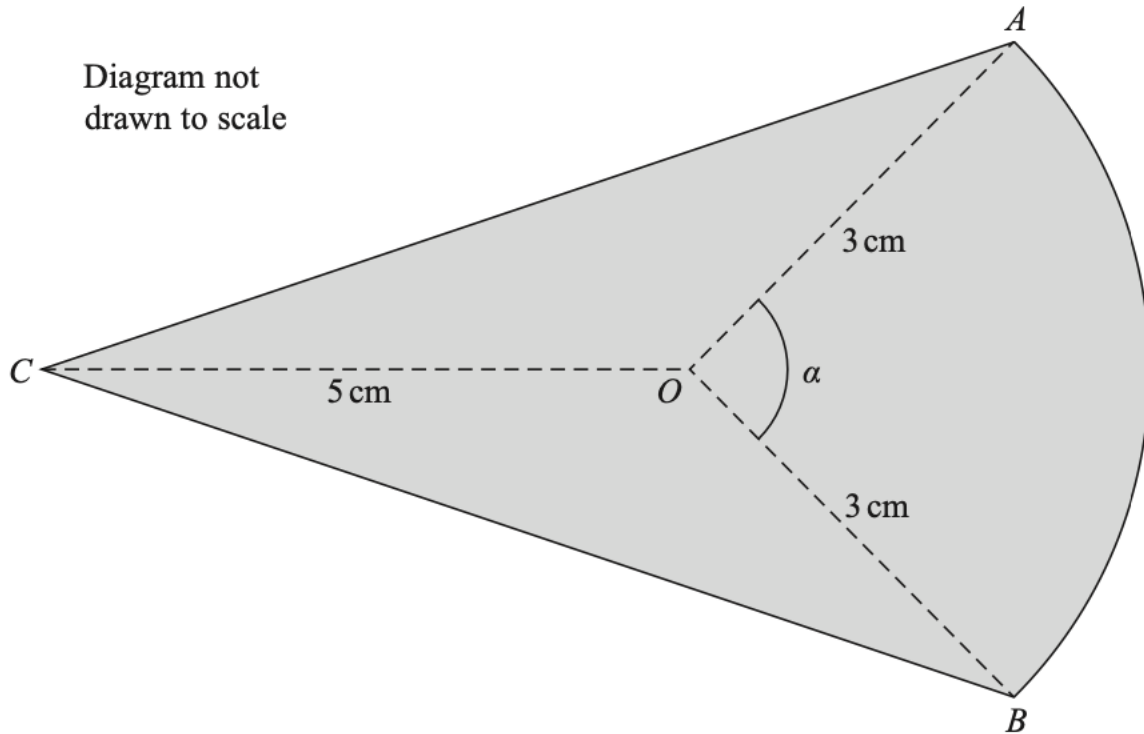


Figure 1

Figure 1 shows the design for a badge.

The design consists of two congruent triangles, AOC and BOC , joined to a sector AOB of a circle centre O .

- Angle $AOB = \alpha$
- $AO = OB = 3 \text{ cm}$
- $OC = 5 \text{ cm}$

Given that the area of sector AOB is 7.2 cm^2

(a) show that $\alpha = 1.6$ radians.

(2)

(b) Hence find

- (i) the area of the badge, giving your answer in cm^2 to 2 significant figures,
- (ii) the perimeter of the badge, giving your answer in cm to one decimal place.

(8)

5.

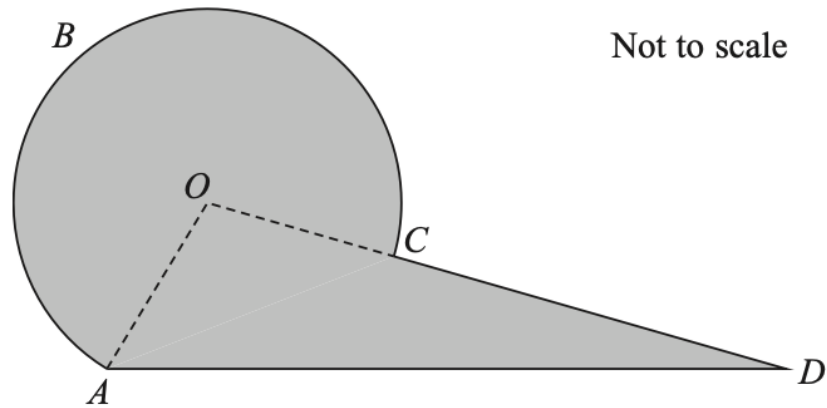


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD .

Given that

- $OA = OC = 6$ m
- $AD = 14$ m
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

find

- (a) the size of angle AOD , in radians, to 3 decimal places, (3)
- (b) the length of arc ABC , in metres, to one decimal place, (2)
- (c) the total area of the viewing platform, in m^2 , to one decimal place. (4)

2021 June Q7 P1

7.

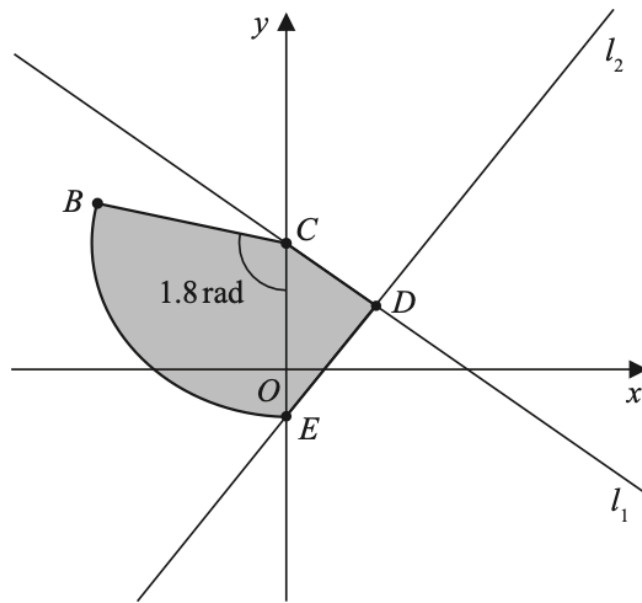


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

(a) State the y coordinate of C .

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE .

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)

7.

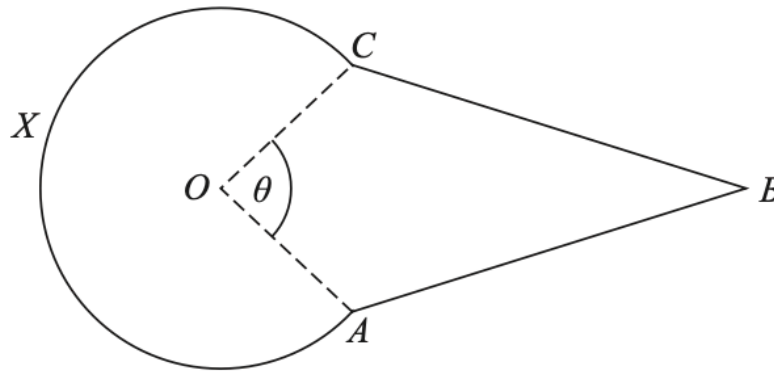


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

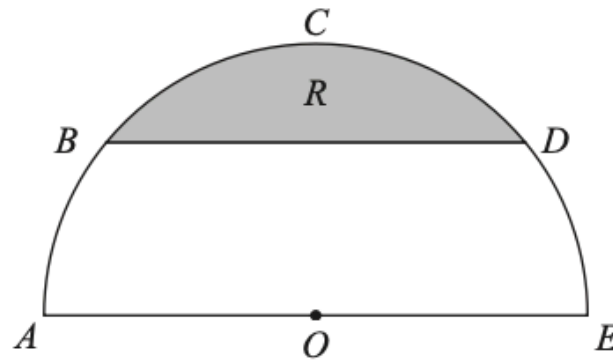
In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

- (a) show that $\theta = 1.64$ radians to 3 significant figures, (4)
- (b) find the perimeter of the sign, in metres to 2 significant figures, (2)
- (c) find the area of the sign, in m^2 to 2 significant figures. (4)

5.



Not to scale

Figure 2

Figure 2 shows a plan view of a semicircular garden $ABCDEOA$

The semicircle has

- centre O
- diameter AOE
- radius 3 m

The straight line BD is parallel to AE and angle BOA is 0.7 radians.

(a) Show that, to 4 significant figures, angle BOD is 1.742 radians.

(1)

The flowerbed R , shown shaded in Figure 2, is bounded by BD and the arc BCD .

(b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

(3)

(c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)

8.

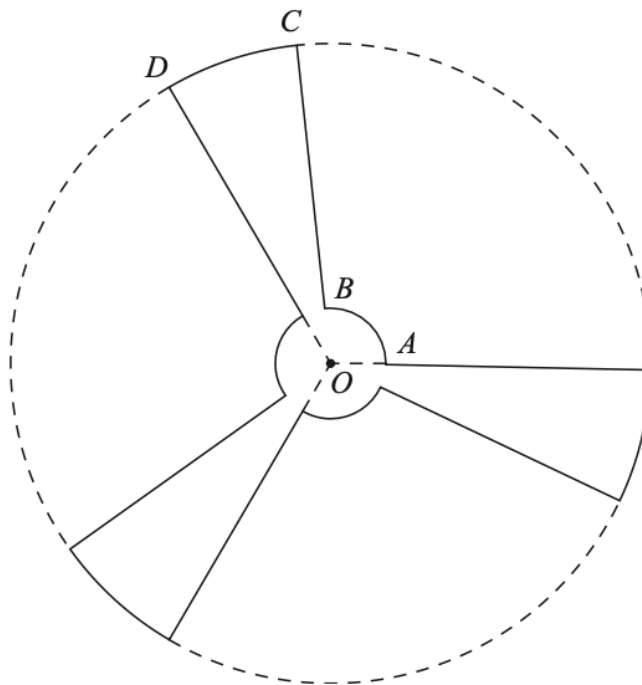


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCO$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

(a) show that the length of the arc CD is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)

(c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

8.

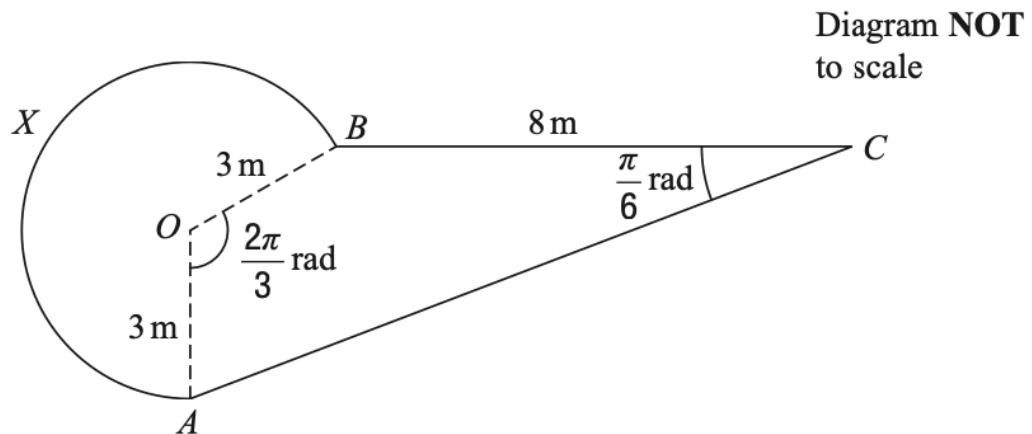


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

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6.

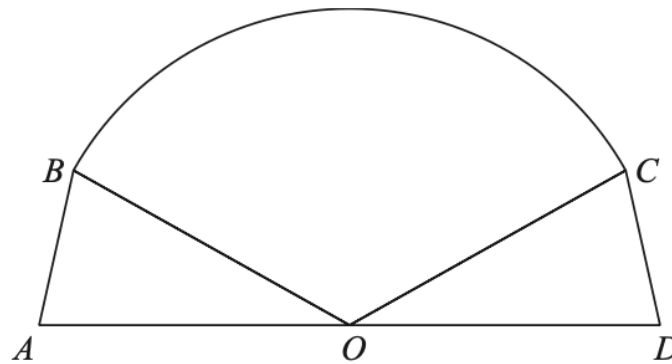


Diagram **NOT**
accurately drawn

Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector BOC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m , to 2 decimal places,

(2)

(b) find the size of angle AOB , in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m , to one decimal place.

(4)

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5.

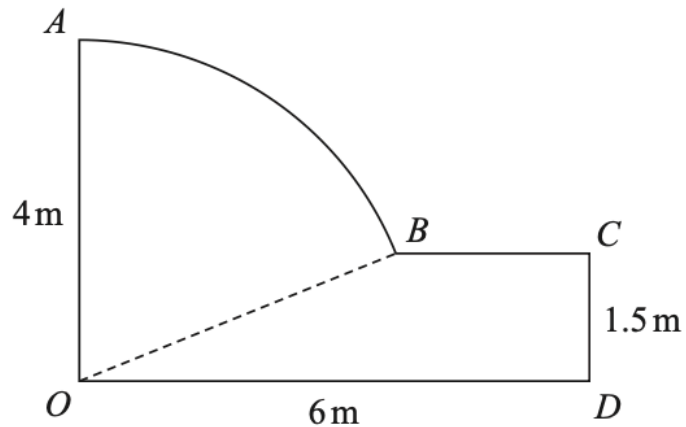


Figure 1

Figure 1 shows the plan for a garden.

In the plan

- OA and CD are perpendicular to OD
- AB is an arc of the circle with centre O and radius 4 metres
- BC is parallel to OD
- OD is 6 metres, OA is 4 metres and CD is 1.5 metres

(a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

9.

Diagram **NOT**
accurately drawn

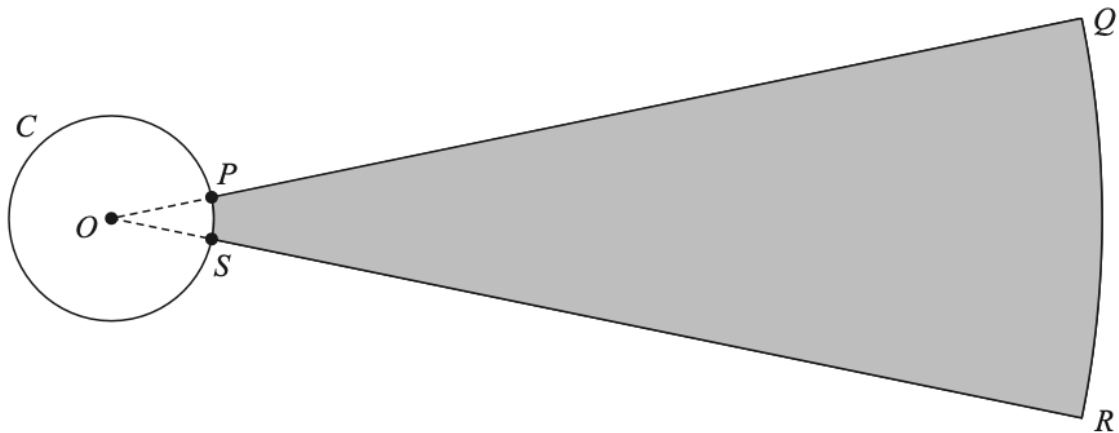


Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2\text{ m}$

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

8.

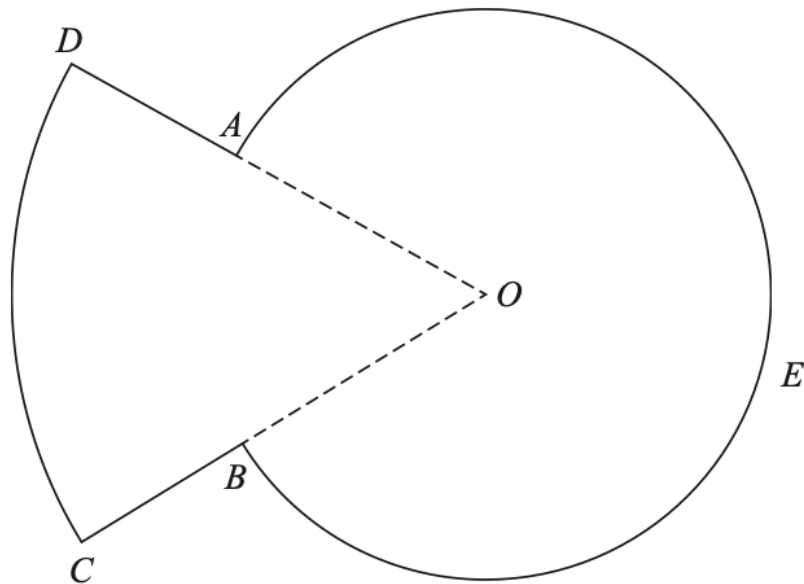


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

5.

Diagram **NOT**
accurately drawn

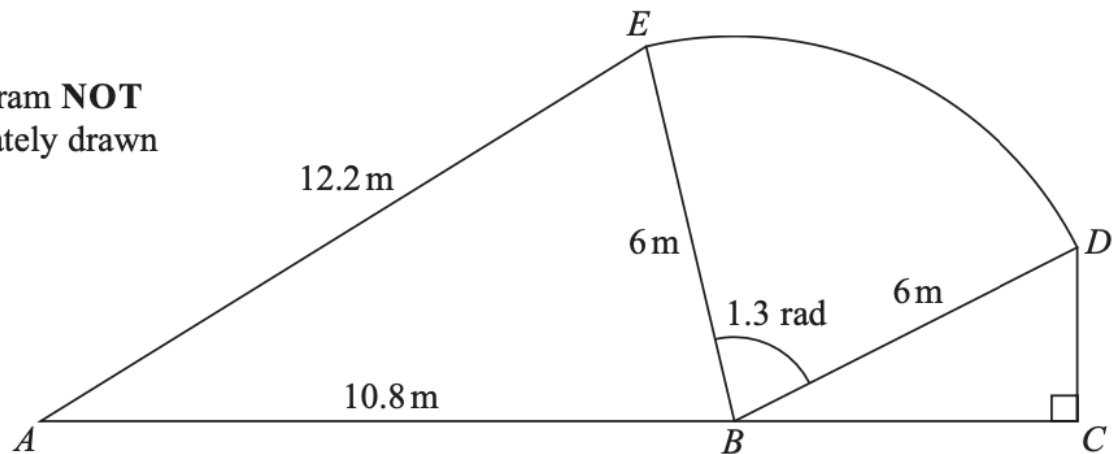


Figure 2

Figure 2 shows the plan view of a garden.

The shape of the garden $ABCDEA$ consists of a triangle ABE and a right-angled triangle BCD joined to a sector BDE of a circle with radius 6 m and centre B .

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- Find the area of the sector BDE , giving your answer in m^2 (2)
- Find the size of angle ABE , giving your answer in radians to 2 decimal places. (2)
- Find the area of the garden, giving your answer in m^2 to 3 significant figures. (3)

6.

Diagram not
drawn to scale

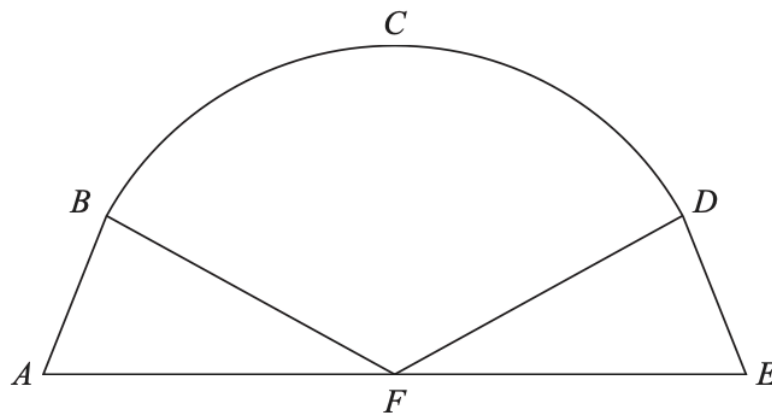


Figure 1

Figure 1 shows a sketch of the entrance to a tunnel.

The shape of the entrance consists of a sector $BCDF$, of a circle centre F , joined to two congruent (identical) triangles ABF and EDF .

Given that

AFE is a straight line

$$AF = FE = 6.4 \text{ m}$$

$$FB = FD = 6.2 \text{ m}$$

$$\text{angle } BFD = 2.275 \text{ radians}$$

- (a) Show that angle $AFB = 0.433$ radians to 3 decimal places. (1)
- (b) Find the perimeter of the entrance to the tunnel, $ABCDEF$, in metres, to one decimal place. (4)
- (c) Find the cross-sectional area of the entrance to the tunnel, $ABCDEF$, in m^2 , to one decimal place. (4)